NAAMA SKITAL

SOFTWARE ENGINEER | PYTHON | C++ | AI/ML

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github.com/naamaskital · linkedin.com/in/naama-skital · Portfolio

PROFILE

Computer Science graduate (Honors Program, GPA 92.17) with software engineering experience in Python/Linux sensor systems and hands-on academic projects spanning machine learning, C++, Java, and data analysis. Enjoys understanding complex systems, solving technical problems, and turning research ideas into reliable implementations.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL **Core:** Algorithms, Data Structures, OOP, Debugging, Testing
AI / Data: PyTorch, scikit-learn, NumPy, pandas, CLIP, NLP, Computer Vision, Model Evaluation
Systems / Tools: Linux, Git, ROS2, DDS, Raspberry Pi, Sensor Synchronization, Jupyter

PROFESSIONAL EXPERIENCE

Software Engineer | LMY *2025–2026*

- Developed Python software for a Linux-based UAV payload collecting LiDAR and navigation sensor data for 3D mapping workflows.
- Worked with ROS2/DDS messaging, sensor timing and synchronization, Raspberry Pi services, and data acquisition pipelines.
- Built validation and diagnostic scripts; investigated asynchronous data, communication delays, and timing inconsistencies.
- Supported post-flight processing and troubleshooting across embedded and ground-side software components.

SELECTED PROJECTS

Multimodal Advertisement Moderation | Python, PyTorch, CLIP, OCR GitHub

- Built and evaluated a four-class image moderation pipeline; compared frozen CLIP, partial fine-tuning, and image/OCR fusion.
- Best fusion model achieved **91.00% accuracy** and **0.6551 macro-F1**; analyzed imbalance and calibration.

Context-Aware Sarcasm Detection | NLP, MiniLM, scikit-learn GitHub

- Compared TF-IDF/logistic regression and sentence-embedding baselines across comment-only and context-aware inputs.
- Achieved **72.53% accuracy** with the comment-only TF-IDF baseline; examined the effect of context.

Topological Data Analysis | Python, GUDHI, Persistent Homology GitHub

- Studied Vietoris–Rips filtrations, H_0/H_1 persistence, diagrams and stability under perturbations on synthetic point clouds.

Software and Algorithms | C++, Java C++ / Java

- Built nine C++ coursework implementations covering OOP, templates, STL and generic containers; implemented Java Bayesian network inference with variable elimination.

EDUCATION

B.Sc. Computer Science | Ariel University *Nov 2023 – Aug 2026*
Honors Program (AI / Data Science) · GPA: **92.17** · Rector's Excellence (first year)

ADDITIONAL EXPERIENCE

Teaching and National Service: Mathematics instruction for at-risk youth; two years of national service in educational settings. Developed communication, mentoring, and collaborative problem-solving skills.